

Multi Collective: Heuristic vs. MILP

Exact validation with absolute Avg JCT, makespan, and solver runtime.

Tenants	Heur. Avg JCT (s)	ILP Avg JCT (s)	Avg JCT Gap (%)	Heur. Makespan (s)	ILP Makespan (s)	Makespan Gap (%)	Heur. Runtime (s)	ILP Runtime (s)
2	0.240297	0.240297	0.0000	0.240297	0.240297	0.0000	1.073	2730.7
3	0.240292	0.240292	0.0000	0.248334	0.248334	0.0000	0.073	812.1
4	0.220135	0.220135	0.0000	0.232217	0.232217	0.0000	0.012	103.6

Absolute values; no normalization. Gap = (Heuristic - ILP) / ILP.